

Supplementary Material for WeatherReasonSeg: A Benchmark for Weather-Aware Reasoning Segmentation in Visual Language Models

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1 Weather Synthesis with Details

We introduce physically interpretable weather synthesis mechanisms in WeatherReasonSeg to systematically model degradation on the original images of ReasonSeg. Low-level image restoration has long studied visual degradation modeling and recovery under challenging imaging conditions [5]. Our design follows two key principles: (1) the synthesis process should approximate real-world imaging mechanisms under adverse weather, rather than relying on heuristic noise injection; and (2) the degradation severity should be continuously controllable to enable quantitative analysis across different weather types and intensity levels.

Based on these principles, we construct physically consistent synthetic data with different severity levels by modeling three representative weather conditions: rain, fog, and snow.

1.1 Adverse Weather Synthesis

Rain Synthesis Rain degradation mainly manifests as two visual effects: sparse rain streaks and accumulation effects caused by dense overlapping streaks. To model these characteristics, we adopt a region dependent rain model and further incorporate rain accumulation [6].

The basic rain model is formulated as:

$$O = B + SR, \quad (1)$$

where O denotes the observed rainy image, B is the clean background image, S represents the rain streak intensity layer, and $R \in \{0, 1\}$ is a region dependent binary mask indicating rain affected pixels. This formulation decouples the spatial

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distribution of rain streaks from their intensity, enabling structured modeling of rain patterns.

The above model mainly describes sparse rain conditions. Under heavy rainfall, dense streaks overlap spatially and temporally, producing accumulation effects that resemble thin fog and reduce scene contrast. To simulate this phenomenon, we introduce a rain accumulation model:

$$O = \alpha \left(B + \sum_{t=1}^s \tilde{S}_t R \right) + (1 - \alpha) A, \quad (2)$$

where $\tilde{S}_t$ denotes rain streak groups with similar orientation and shape, s is the number of rain groups, A is the global atmospheric light, and α represents the transmission factor controlling haze like attenuation.

By adjusting the number of rain groups s , overlap levels, and transmission α , we generate a continuous degradation spectrum from light rain to heavy rainfall.

Fog Synthesis Fog degradation mainly arises from atmospheric scattering. Under the assumption of homogeneous fog and stable illumination, the imaging process can be approximated by the atmospheric scattering model [8]:

$$I(x) = J(x)t(x) + A(1 - t(x)), \quad (3)$$

where $I(x)$ is the observed foggy image, $J(x)$ is the clear background image, A denotes the global atmospheric light, and $t(x)$ represents the scene transmission map.

The transmission is related to the scene depth $d(x)$ and scattering coefficient β :

$$t(x) = \exp(-\beta d(x)). \quad (4)$$

As the depth increases, the direct component $J(x)t(x)$ attenuates while the atmospheric light component $A(1 - t(x))$ becomes dominant, resulting in reduced contrast and degraded visibility.

In practice, the haze free image is first converted to linear RGB space, the fog image is synthesized using the above equations, and the result is converted back to sRGB to preserve realistic brightness distribution.

Fog severity is controlled by the scattering coefficient β , which is determined from meteorological visibility R_m :

$$\beta = -\frac{\ln(\epsilon)}{R_m}, \quad (5)$$

where ϵ denotes the perceptual contrast threshold. By varying R_m , we generate fog conditions ranging from light to dense fog.

Snow Synthesis Snow degradation mainly results from particle occlusion, light scattering, and contrast reduction under dense snowfall. Compared with rain streaks, snow particles appear as irregular high intensity granular structures with varying scales and non-uniform spatial distributions.

We model snowy degradation as the combination of background imaging, snow particle layers, and atmospheric scattering [3]:

$$I(x) = \gamma J(x) + \sum_{k=1}^n S_k(x; \rho, \sigma), \quad (6)$$

where $I(x)$ denotes the synthesized snowy image, $J(x)$ is the clean background image, and $S_k(x; \rho, \sigma)$ represents the k -th snow particle layer generated with particle density ρ and particle scale distribution σ . The parameter γ denotes the transmission coefficient that models the visibility attenuation caused by snowfall. By adjusting ρ , σ , and γ , different snowfall severities ranging from light snow to heavy snow can be simulated.

Each snow layer consists of randomly generated particles with varying scale, brightness, and transparency, producing localized occlusions while preserving partial background visibility. Snowfall severity is controlled by the number of particle layers n , particle scale distribution, and transmission attenuation $\gamma(x)$, enabling gradual degradation from light snowfall to heavy snow conditions.

1.2 Adverse Weather Severity Parameter Settings

To simulate different levels of environmental degradation, we control the severity of fog, rain, and snow through a set of physically interpretable parameters. The specific configurations for light, moderate, and heavy conditions are summarized in Table 1. For fog synthesis, A denotes the global atmospheric light, while β represents the scattering coefficient that determines the strength of atmospheric attenuation. As β increases, light scattering becomes stronger, leading to reduced visibility and more severe fog degradation. For rain synthesis, s denotes the number of rain streak groups, which directly controls the density of rain streaks, while α represents the transmission factor that models the accumulation effect of dense rainfall. Larger s values produce denser rain streaks, and smaller α values correspond to stronger attenuation, resulting in heavier rain appearance. For snow synthesis, ρ denotes the particle density controlling the number of snow particles, σ represents the particle scale distribution that determines the size variation of snowflakes, and γ denotes the transmission coefficient that models visibility attenuation under snowfall. Increasing ρ and σ leads to denser and larger snow particles, while decreasing γ further reduces scene contrast, producing progressively heavier snowfall effects.

Table 1. Parameter settings for different adverse weather severities.

Weather	Light	Moderate	Heavy
Fog	$A=0.8, \beta=0.32$	$A=0.8, \beta=1.2$	$A=0.8, \beta=2.0$
Rain	$s=200, \alpha=0.9$	$s=700, \alpha=0.85$	$s=1000, \alpha=0.8$
Snow	$\rho=0.50, \sigma=3.0, \gamma=0.80$	$\rho=0.85, \sigma=4.0, \gamma=0.70$	$\rho=0.90, \sigma=4.5, \gamma=0.55$

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messages = [{"role": "system", "content": "You are an AI visual assistant observing an RGB image. You are also provided
with a mask image and the category label of the target object in the scene.
Your task is to: (1)Analyze the RGB image and identify the target object corresponding to the mask (i.e., the object should be
visually recognizable and clearly separable from the background). (2)Generate five natural language questions from five
perspectives: functional positioning, application scenarios, structural features, interaction relationships, and requirement
matching. (3) All questions should clearly refer to the same target object.Each question must be answerable solely by observing
the RGB image. (4) Under no circumstances should the questions mention or reference segmentation masks. Maintain a natural
conversational tone while satisfying the length requirements, and avoid overly rigid wording.
"},
{
  "role": "user",
  "content": [
    {"type": "text", "text": f'Class label (from segmentation): {class_name}. Here is an RGB image and a binary mask
for that class. Generate 5 questions.'},
    {"type": "image_url", "image_url": {"url": f'data:image/{rgb_fmt};base64,{rgb_b64}' }},
    {"type": "image_url", "image_url": {"url": f'data:image/{mask_fmt};base64,{mask_b64}' }}
  ],
},
]

```

Fig. 1. Prompt construction for reasoning query generation. The prompt consists of a system instruction defining the reasoning task and a user message containing the class label, RGB image, and corresponding binary mask to guide the generation of five reasoning oriented questions.

2 Data Annotation of WeatherReasonSeg

2.1 Dataset Construction Pipeline

To construct the WeatherReasonSeg benchmark dataset, we propose an efficient semi-automatic data construction pipeline. Each sample in the dataset consists of an image, five reasoning queries, and a target binary mask. The overall process is described as follows.

Step 1: Data Preparation and Pre-extraction We first manually annotate 200 samples, covering queries and masks across all reasoning dimensions. Using a mask based extraction module, each image is processed at the pixel level to automatically decompose the scene into object masks and their corresponding descriptive labels.

Step 2: Prompt Engineering with Large Language Models The extracted descriptive labels are incorporated as soft prompts, together with the original image masks and manually annotated example samples, to construct the complete prompt input for the Qwen large language model. Through an in context learning mechanism, the model generates five candidate reasoning queries for each sample.

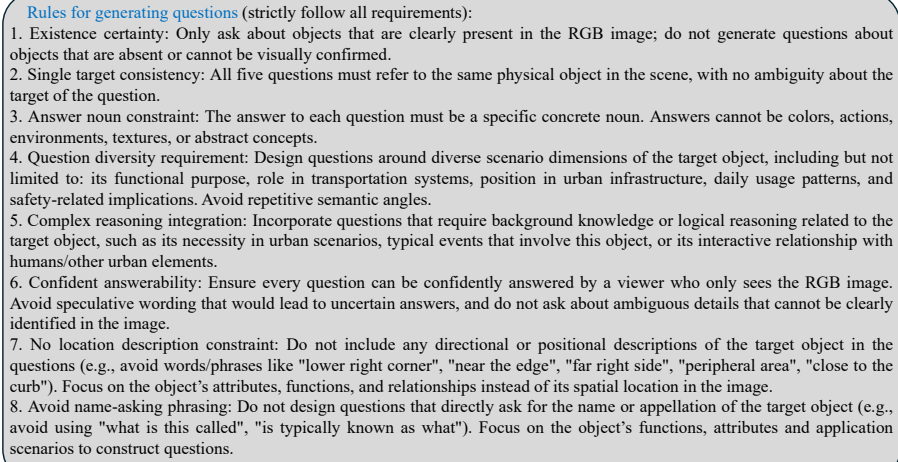


Fig. 2. The figure outlines the verification rules used to ensure that generated questions satisfy object existence certainty, single target consistency, noun-based answers, reasoning diversity, and image grounded answer ability.

Step 3: Verification and Iteration A dual stage filtering mechanism combining human experts and large language models is applied to verify the generated queries. Candidate queries are screened according to predefined rules. If semantic ambiguity or logical inconsistency is detected, the queries are either rewritten by the LLM or manually corrected by experts until high quality reasoning queries are obtained.

The prompt construction process in Step 2 is illustrated in Figure 1, while the rule based filtering procedure used for verification in Step 3 is illustrated in Figure 2.

2.2 Annotation Quality Control and Reliability

Our data construction pipeline uses an LLM to generate reasoning queries and their corresponding five-dimension labels, while human participants serve as auditors rather than primary annotators. Specifically, we recruited 10 auditors with computer vision backgrounds to inspect the generated samples from three aspects: (i) semantic consistency between the query and the target mask, (ii) potential query ambiguity, and (iii) whether the query can be executed as a valid segmentation instruction. To ensure reliable quality control, we adopt a three-stage progressive sampling protocol. We first construct a seed generation pool, then expand it to a 10K pilot set, and finally generate the full dataset. At each stage, 300 samples are randomly inspected. If the acceptance rate falls below 95%, the current generation pool is discarded, the prompt is refined, and the samples are regenerated.

Since the dimension labels are produced by the same LLM under a unified prompting scheme, traditional inter-annotator agreement for independent dimension assignment is not directly applicable. Instead, human auditors verify whether each generated query is consistent with its intended reasoning dimension and whether all five queries refer to the same target object. Samples with semantic ambiguity, dimension mismatch, or inconsistent target references are either corrected or regenerated.

3 Implementation details

We adopt 8 NVIDIA 24G 3090 GPUs for inference. We evaluate four representative frameworks on WeatherReasonSeg: Grounded-SAM [4], which is grounding-based, and three reasoning-based models: LISA [1], Seg-R1 [7], and Seg-Zero [2]. Specifically, LISA adopts a vision-language backbone based on LLaVA-1.5, while Seg-R1 and Seg-Zero are built upon the Qwen2.5-VL architecture. For fairness, all models are evaluated using their official implementations and default inference configurations. During evaluation, the input image and the corresponding query are provided to each framework to predict the segmentation mask of the target object.

4 More experiments

4.1 Qualitative Analysis Across Reasoning Dimensions

Figure 3 shows representative qualitative examples from our benchmark under real-world conditions, illustrating how reasoning queries from different semantic dimensions affect VLM reasoning segmentation. In this example, the target is a service vehicle, and queries are constructed from five perspectives: Function, Application, Structure, Relationship, and Requirement. The model correctly identifies the vehicle under Function, Application, Structure, and Relationship queries, indicating that when reasoning cues align with observable visual attributes, such as object category, structural characteristics, or roles in urban logistics, VLMs reasoning segmentation can reliably localize the target. However, the model fails under the Requirement query, confusing the functional roles of objects in the construction scene. Although the query asks for a vehicle suitable for transporting construction materials, the model incorrectly selects the temporary construction barrier. This error suggests that the model relies on contextual co-occurrence cues rather than correctly reasoning about object functions.

Figure 4 further reveals limitations under complex situational contexts. In the rainy highway accident scenario, the model successfully identifies the crashed vehicle when queries describe the accident event or structural characteristics, which provide clear visual grounding. In contrast, the model fails under certain Application and Relationship queries that require higher level situational reasoning, such as identifying the vehicle monitored by police or interpreting emergency

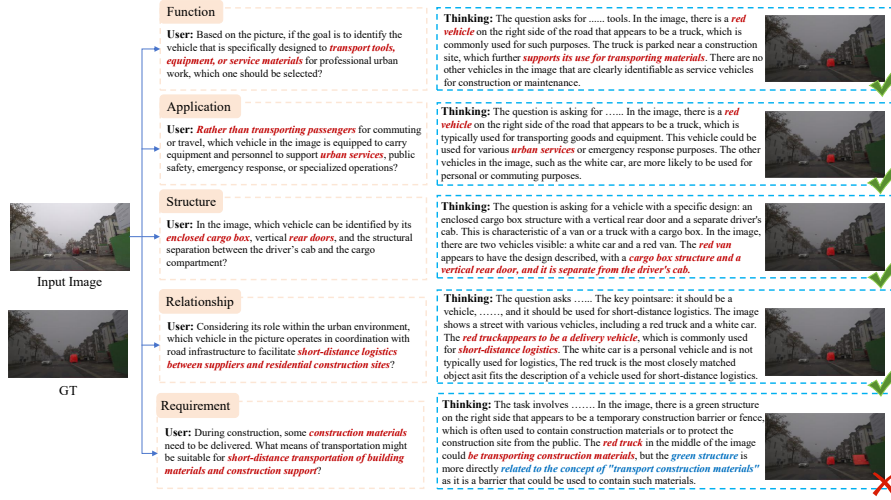


Fig. 3. Given the same input image and target object, we construct queries from five semantic perspectives—Function, Application, Structure, Relationship, and Requirement. The reasoning segmentation method adopts Seg-Zero. The right panel illustrates the reasoning process and segmentation predictions of the VLM model, where green check marks denote correct predictions and the red cross indicates a failure case. This example highlights how different reasoning dimensions influence the model’s ability to correctly localize the target object.

response dynamics. In these cases, the model confuses the crashed vehicle with the police car.

Overall, the performance of reasoning segmentation varies significantly across different reasoning dimensions. Queries grounded in directly observable visual features tend to produce more stable predictions, whereas those requiring contextual reasoning, event interpretation, or implicit task requirements are more prone to failure. This observation is consistent with our quantitative findings and highlights the importance of evaluating reasoning segmentation across diverse semantic perspectives and adverse weather conditions.

4.2 Qualitative Comparison Under Adverse Weather Conditions

Figure 5 shows the performance of reasoning based segmentation models degrades significantly under adverse weather conditions, consistent with the quantitative observations. When environmental degradation interferes with visual cues, reasoning based methods often struggle to accurately identify targets. For example, in rainy or snowy weather, Seg-Zero and Seg-R1 sometimes produce incomplete or off target predictions, while LISA may incorrectly highlight surrounding objects with similar contextual cues. These errors indicate that adverse weather not only affects low level visual perception but also interferes with the

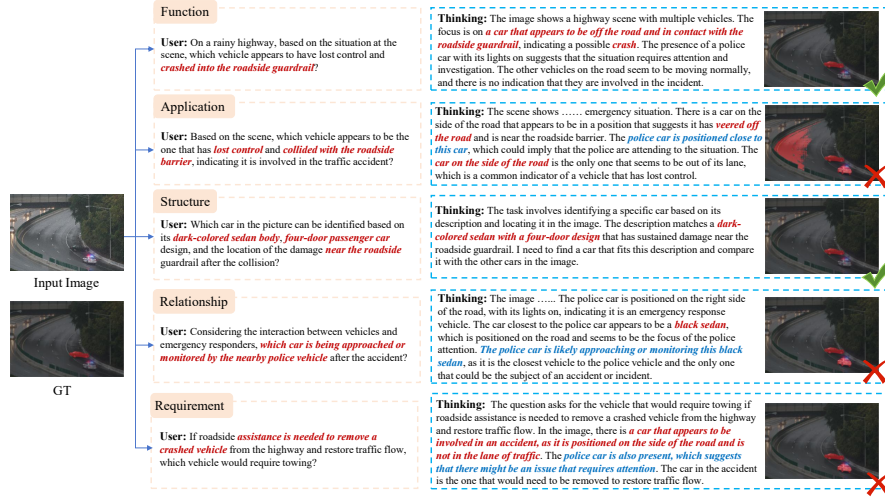


Fig. 4. Given the same input image and target object, we test the reasoning and segmentation prediction results of the VLM model from five queries, where a green check indicates correct prediction and a red cross indicates prediction failure. This example highlights how different reasoning dimensions affect the model’s ability to correctly locate the target object.

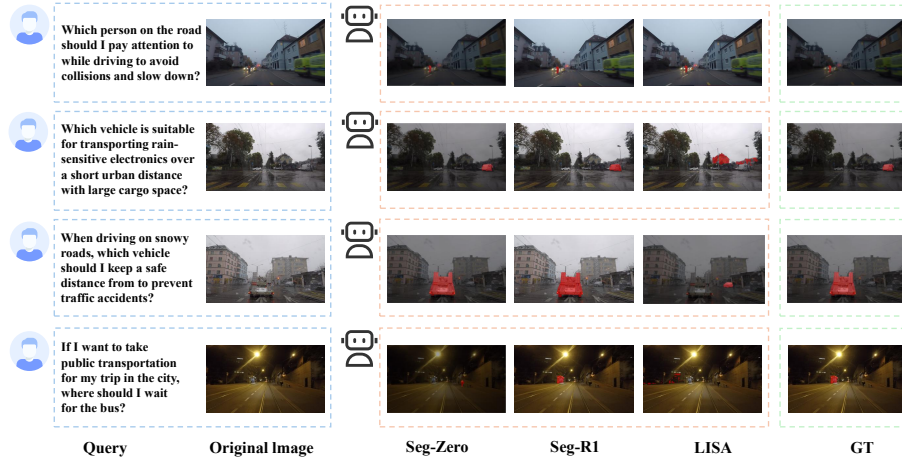


Fig. 5. Qualitative results of reasoning segmentation under adverse weather conditions. For each example, we show the query, the original image, and the segmentation predictions produced by Seg-Zero, Seg-R1, and LISA, together with the ground truth (GT). The results highlight the different reasoning segmentation performance of VLM based methods in environments such as rain, snow, and nighttime.

Table 2. Results on a subset of real-world adverse-weather conditions with closed-source API-based evaluation.

Method	Fog		Rain		Snow		Night	
	gIoU	cIoU	gIoU	cIoU	gIoU	cIoU	gIoU	cIoU
Gemini-2.5-Pro	16.0	23.0	10.1	12.3	10.7	19.4	10.6	12.4
GPT-5.4	23.9	26.1	18.0	19.6	18.5	20.1	18.2	19.9
Qwen3-VL-Plus	40.1	46.6	30.6	41.0	32.6	39.0	23.2	28.7
Seg-Zero-7B	40.6	47.1	35.1	38.7	36.8	41.6	28.9	29.4

high level reasoning processes required for language guided segmentation. In contrast, truth masks demonstrate that accurate localization typically relies on the integration of object level features and contextual understanding, which remains a challenge for current VLM based reasoning based segmentation methods.

4.3 Additional Results with Closed-source Models

In addition to open-source baselines, we further evaluate GPT-5.4, Gemini-2.5-Pro, and Qwen3-VL-Plus on a subset of WeatherReasonSeg. The detailed results are presented in Tab. 2. These experiments provide additional evidence for analyzing whether stronger closed-source VLMs can alleviate the performance degradation under adverse weather conditions.

4.4 Additional Results under Non-weather Corruptions

To verify that our analysis is not restricted to weather-induced degradation, we additionally evaluate representative in-camera corruptions, including Gaussian noise, blur, and JPEG compression. As shown in Tab. 3, reasoning segmentation models exhibit clear performance drops under these corruptions, suggesting that our analytical framework can generalize beyond weather-specific scenarios. Furthermore, compared with simple corruption injection, physics-aware weather synthesis produces structured and scene-dependent degradations, making our benchmark more realistic and challenging.

Table 3. Performance under severe in-camera corruptions.

Method	Gaussian Noise				Blur				JPEG			
	val		test		val		test		val		test	
	gIoU	cIoU	gIoU	cIoU	gIoU	cIoU	gIoU	cIoU	gIoU	cIoU	gIoU	cIoU
LISA-7B	43.1	54.7	41.7	48.1	44.4	54.4	38.7	47.8	42.3	52.5	44.2	47.6
Seg-Zero-7B	55.8	54.0	53.4	52.4	57.6	55.1	47.5	48.0	59.3	56.6	56.9	51.8

5 Limitations

Although WeatherReasonSeg provides a comprehensive benchmark for evaluating VLM reasoning segmentation under adverse weather conditions, it still has several limitations. First, the synthetic weather data are generated using simulation pipelines, which may not fully capture the complex physical properties of real-world weather phenomena. As a result, certain factors such as dynamic precipitation patterns, lighting interactions, and sensor noise are only partially modeled. Second, although our real-world subset includes multiple adverse conditions such as rain, snow, fog, and nighttime scenes, the diversity of environmental conditions and object categories remains limited compared with the vast variety of real-world scenarios. Since the real-world subset is mainly constructed from driving scenes with reliable pixel-level annotations, it may also contain an urban-driving bias, including a relatively high proportion of transportation-related objects. Expanding the dataset to incorporate more physically accurate weather simulations and more diverse domains, such as indoor, aerial, and industrial scenarios, is an important direction for future work.

6 Ethical Statement

The dataset employed in this study contains no personally identifiable information and does not involve human subjects, sensitive personal data, or any protected information. All data were obtained from publicly available sources or provided under appropriate agreements, and are used exclusively for academic research purposes. Data handling procedures ensure that confidentiality and data integrity are maintained throughout the research process. Consequently, this study raises no ethical concerns and poses no risks to individuals or society.

References

1. Lai, X., Tian, Z., Chen, Y., Li, Y., Yuan, Y., Liu, S., Jia, J.: Lisa: Reasoning segmentation via large language model. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 9579–9589 (2024)
2. Liu, Y., Peng, B., Zhong, Z., Yue, Z., Lu, F., Yu, B., Jia, J.: Seg-zero: Reasoning-chain guided segmentation via cognitive reinforcement. arXiv preprint arXiv:2503.06520 (2025)
3. Michaelis, C., Mitzkus, B., Geirhos, R., Rusak, E., Bringmann, O., Ecker, A.S., Bethge, M., Brendel, W.: Benchmarking robustness in object detection: Autonomous driving when winter is coming. arXiv preprint arXiv:1907.07484 (2019)
4. Ren, T., Liu, S., Zeng, A., Lin, J., Li, K., Cao, H., Chen, J., Huang, X., Chen, Y., Yan, F., et al.: Grounded sam: Assembling open-world models for diverse visual tasks. arXiv preprint arXiv:2401.14159 (2024)
5. Tang, Q., Guo, X., Wei, X., Li, X., Wang, D., Zhang, S.: Sat-uir: Self-assessment training for semi-supervised underwater image restoration. IEEE Transactions on Circuits and Systems for Video Technology (2025)

6. Yang, W., Tan, R.T., Feng, J., Liu, J., Guo, Z., Yan, S.: Deep joint rain detection and removal from a single image. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 1357–1366 (2017)
7. You, Z., Wu, Z.: Seg-r1: Segmentation can be surprisingly simple with reinforcement learning. arXiv preprint arXiv:2506.22624 (2025)
8. Zhang, Y., Ding, L., Sharma, G.: Hazerd: an outdoor scene dataset and benchmark for single image dehazing. In: 2017 IEEE international conference on image processing (ICIP). pp. 3205–3209. IEEE (2017)